ABU DHABI UNIVERSITY
College of Business
MSc in Strategic Digital Transformation

From Traffic Claims to Welfare Evidence

A data-gated quasi-experimental evaluation of smart-traffic investments in Dubai

Research Proposal for the Master of Science in Strategic Digital Transformation

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Submitted: 28 May 2026

Administrative Information

Table 1. Administrative information for the proposal submission.

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Submission cycle	Fall 2026 application cycle
Date	17 May 2026

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Acronyms

Table 2. Acronyms used in the proposal.

Acronym	Definition
AED	United Arab Emirates dirham
ADU	Abu Dhabi University
ATT	Average treatment effect on the treated
BCR	Benefit-cost ratio
CBA	Cost-benefit analysis
DiD	Difference-in-differences
DITSC	Dubai Intelligent Traffic Systems Centre
DLD	Dubai Land Department
ITS	Intelligent transport systems
MCDA	Multi-criteria decision analysis
MDE	Minimum detectable effect
MOI	Ministry of Interior
NPV	Net present value
QGIS	Quantum Geographic Information System
RTA	Roads and Transport Authority
SUMO	Simulation of Urban Mobility
UTC-UX	Urban Traffic Control user-experience traffic-signal system

1. Abstract and Keywords

1.1 Abstract

Dubai is no longer at the stage of pilot rhetoric in intelligent traffic management. DITSC was announced as a AED 590 million network upgrade, expanding ITS coverage from 11 percent to 60 percent of the road network. In 2025, RTA added UTC-UX Fusion, an AI- and digital-twin-based signal-control system, and later publicised first-phase flow gains of 16 percent to 37 percent. Those figures matter politically, but they do not by themselves establish a counterfactual traffic effect or a net welfare return.

That gap between project claim and welfare evidence drives this proposal. I ask whether exposure to DITSC or UTC-UX is associated with lower delay, lower speed loss, or more reliable travel times in Dubai corridors, and whether any observed gains survive cost-benefit appraisal once capital cost, operating cost, time savings, accidents, emissions, and uncertainty are valued in the same framework.

The empirical strategy is conditional rather than rhetorical. If treatment timing, geography, controls, and outcomes can be reconstructed at corridor or intersection level, I will estimate traffic effects with difference-in-differences and event-study models. If that panel cannot be built, I will not stretch weaker evidence into a causal claim. The design then steps down to property-capitalisation evidence from Dubai Land Department transactions, and finally to scenario-based appraisal where effects are assumed rather than observed.

The study is meant to deliver a reproducible evaluation protocol for smart-traffic investment under restricted agency data. It separates three tasks that are often blurred in policy discussion: estimating causal traffic effects, reading secondary real-estate evidence, and running scenario appraisal. That separation is the main safeguard against saying more than a master's thesis can defend.

Table 3. Keyword structure for the study by category.

Category	Keywords
Intervention	DITSC; UTC-UX Fusion; intelligent traffic systems;

Category	Keywords
	adaptive signal control; digital twin traffic management
Method	difference-in-differences; event study; synthetic control; hedonic pricing; cost-benefit analysis
Tools	QGIS; SUMO; treatment registry; spatial buffers; reproducible data dictionary
Outcomes	delay; speed loss; travel-time reliability; accidents; emissions; welfare return; benefit-cost ratio
Context	Dubai RTA; UAE smart mobility; transport policy evaluation; urban infrastructure appraisal

2. Introduction and Research Problem

2.1 A quantified problem before the research agenda

Dubai's smart-traffic programme is large enough to warrant economic evaluation, not just descriptive reporting. DITSC alone was announced at AED 590 million as a network-wide upgrade in incident detection, monitoring, and traffic control. UTC-UX added a second layer: AI-assisted signal optimisation, predictive analytics, and digital-twin control at major intersections. RTA later reported flow improvements of 16 percent to 37 percent on selected corridors. Those figures show policy scale. They do not yet show welfare return.

A reported flow gain is not an average treatment effect. It says nothing, by itself, about the counterfactual corridor, rollout timing, spillovers to nearby roads, pre-existing trends, or whether the benefit survives CAPEX and OPEX. I therefore start from a narrower economic question: when public agencies deploy named smart-traffic systems, how far can a researcher move from project claims to defensible welfare appraisal?

Dubai is a hard but useful setting for that question. It combines a capable transport authority, a rapidly evolving digital traffic programme, tolling data that capture demand, and a large property market. Those same strengths also make identification fragile. If rollout becomes system-wide too quickly, untreated controls disappear. If agency microdata remain closed, the admissible claim must contract.

Table 4. Scientific gap motivating the study.

Scientific gap
The missing knowledge is not whether Dubai deploys smart-traffic technology. The missing knowledge is whether named smart-traffic treatments produce welfare returns that survive counterfactual testing, cost valuation, and transparent downgrading when the data are incomplete.

2.2 Why the timing matters

Timing matters here for empirical rather than cosmetic reasons. UTC-UX sits in a recent 2025–2026 rollout window. That creates a narrow opportunity in which staggered adoption, partial exposure, or later-treated units may still support comparison. The window can close quickly. Once most major intersections are treated, a conventional untreated-control design becomes difficult to defend.

That is why the first twelve weeks are not a generic literature-review phase. They are the feasibility screen. In that period I will audit sources, assemble a treatment registry, test DLD extraction, map candidate exposure zones, and decide whether the study can sustain Level 1 causal claims or must step down to Level 2 or Level 3 evidence.

2.3 GIFT positioning

The proposal's positioning is stronger when the logic is visible as a single strategy map rather than as an isolated table.

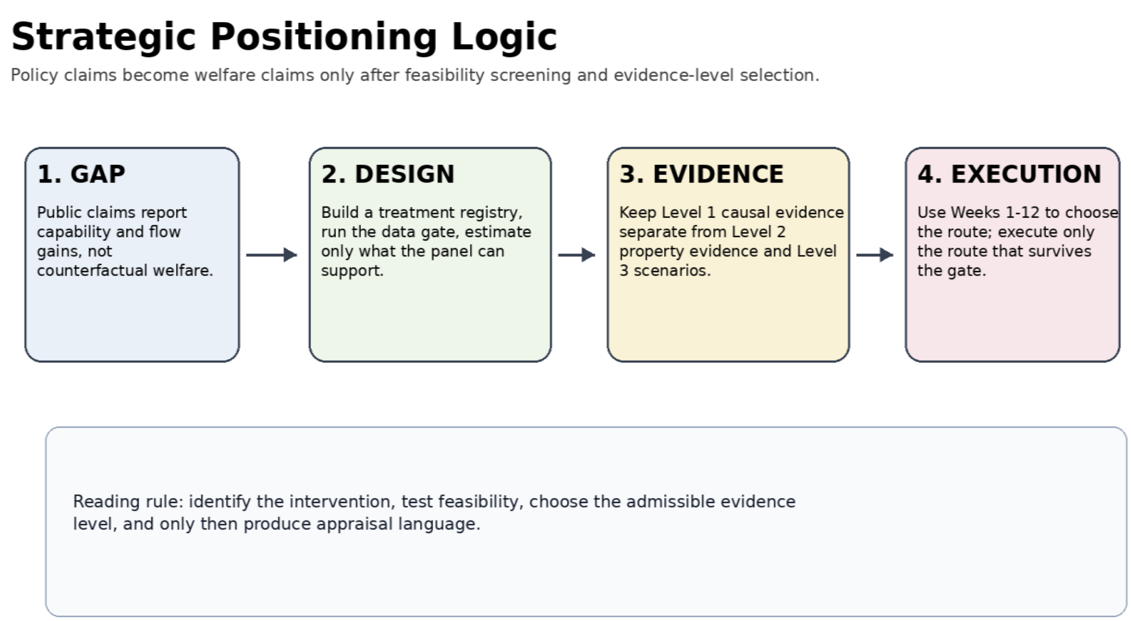

Figure 1. Strategic positioning logic showing how public policy claims are converted into admissible welfare claims only after feasibility screening and evidence-level selection.

Table 5. GIFT positioning summary for the proposal.

GIFT element	Positioning
Gap	Public communication reports system capabilities and selected flow gains, but it does not establish corridor-level counterfactual effects or net welfare return.
Idea	Build a treatment registry, run a data gate, estimate traffic effects only if the panel supports them, and monetise observed or assumed effects with strict claim labels.
Feasibility	Feasible as an MSc thesis only if evidence levels remain explicit: Level 1 for corridor causal evidence, Level 2 for property-capitalisation evidence, Level 3 for scenario appraisal.

GIFT element	Positioning
Timeline	Weeks 1-12 decide the empirical route. Months 4-12 execute the route that survives the data gate.

2.4 Why Dubai, why me, why ADU

Table 6. Rationale for Dubai, researcher fit, timing, and ADU alignment.

Question	Answer
Why Dubai?	Dubai offers a rare combination: named smart-traffic treatments, fast deployment, public performance claims, toll-demand signals, real-estate transaction data, and policy pressure to appraise digital mobility investment.
Why me?	My Casablanca project already combined traffic exposure, QGIS/SUMO thinking, ROI, CAPEX/OPEX logic, and spatial investment evaluation. This proposal turns that applied experience into a stricter economics-and-policy design.
Why now?	The 2025-2026 UTC-UX rollout is recent enough to create a possible evaluation window, but broad deployment may soon erase clean controls.
Why ADU?	ADU's MSc in Strategic Digital Transformation offers the digital transformation, data-analysis, and applied project environment needed to convert a transport-technology question into an appraisal and identification problem. Faculty alignment will be verified before submission; no supervisory commitment is assumed here.

3. Research Questions, Objectives, and Contribution

3.1 Central question

Under verifiable treatment timing and geography, do DITSC and UTC-UX exposure reduce traffic frictions in Dubai corridors, and how should welfare claims be downgraded when the data only support weaker evidence?

3.2 Sub-questions

4. Did smart-traffic exposure reduce delay, speed loss, or travel-time unreliability relative to credible controls?
5. When traffic effects can be estimated, do monetised benefits exceed capital and operating costs under transparent CBA assumptions?
6. If traffic microdata are not available, do nearby property transactions show accessibility capitalisation around treated buffers?
7. How should decision-makers read positive, null, weak, or non-causal results when the evidence level changes?

7.1 Operational objectives

Table 7. Operational objectives, deadlines, and acceptance criteria for the proposed study.

Objective	Task	Deadline	Acceptance criterion
O1	Build a DITSC/UTC-UX treatment registry with source URL, date precision, geography precision, and exposure level.	End of Month 1	Registry v0 separates Level 1, Level 2, and Level 3 units.
O2	Complete a Data Gate Memo.	End of Month 3	Memo chooses confirmatory DiD, exploratory DiD, hedonic fallback, or scenario appraisal.
O3	Construct the primary panel or fallback dataset.	End of Month 6	Dataset contains unit-period structure, codebook, missingness log, and claim status.
O4	Estimate models only if diagnostics permit.	End of Month 8	Output includes matching balance, pre-trend checks, robustness, and claim classification.
O5	Translate eligible effects into NPV and BCR.	End of Month 10	CBA workbook reports low, central, and high scenarios and labels observed versus assumed inputs.
O6	Prepare a claim-aware policy	End of Month 11	The matrix reports rankings

Objective	Task	Deadline	Acceptance criterion
	matrix.		together with uncertainty and evidence level.

7.2 Hypotheses and decision tests

Table 8. Testable statements, appraisal tests, and claim-discipline rules.

Label	Statement	Decision rule
H1 - traffic effect	DITSC/UTC-UX exposure reduces average delay, speed loss, or travel-time variability by a policy-relevant margin relative to matched controls.	Refuted or downgraded if the estimate is zero, wrongly signed, unstable, or fails pre-trend and balance diagnostics.
D1 - CBA test	Under observed Level 1 effects, BCR exceeds 1 only if monetised time, reliability, safety, or emissions benefits dominate CAPEX and OPEX.	This is an appraisal test, not a causal hypothesis.
H2 - property capitalisation	Properties near treated corridors or intersections show accessibility capitalisation relative to comparable untreated locations.	Refuted if the buffer-post interaction is null, unstable, or absorbed by confounders.
R1 - null-result rule	Weak or null results mean the available data do not verify welfare returns; they do not prove that the technology failed.	This rule prevents overclaiming.

Notes: BCR = benefit-cost ratio. D1 is an appraisal test rather than a causal hypothesis; R1 is a rule for interpreting null or weak results.

7.3 Contribution

I am not presenting this thesis as a solution to Dubai congestion. Its contribution is narrower and, in policy terms, more useful: a treatment registry, a data-gated evaluation protocol, a welfare-accounting workbook, and a claim-aware policy matrix. Put differently, the study asks what can still be learned credibly when agencies publish project claims but do not necessarily release the microdata needed for clean causal inference.

8. Literature Review: From Smart-Traffic Claims to the Welfare-Evidence Gap

8.1 Smart traffic and transport performance

The smart-mobility literature is useful here, but only up to a point. Butler, Yigitcanlar, and Paz (2020) map the operational channels through which intelligent transport systems can affect congestion, emissions, safety, and accessibility. Vu and Preston (2022) show how transport options can be compared through social-cost logic and simulation. I use this literature to define outcome categories and mechanisms, not as evidence that Dubai's named interventions generated welfare gains.

8.2 Smart-city policy and productivity

The broader smart-city literature offers methodological cues rather than direct proof for this case. Fan, Zhang, and Su (2023) and Chen (2022) show that digital urban policy can affect productivity and environmental performance in some settings. But their policy environments, selection processes, and outcome scales are not the same as Dubai's corridor-level traffic problem. I therefore treat them as evidence that quasi-experimental urban digital-policy work is possible, not as evidence that Dubai's systems worked.

8.3 Causal identification under staggered rollout

Difference-in-differences is useful only when treatment timing is observed and untreated or later-treated units provide a credible counterfactual. That is why the recent DiD literature matters here. Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Abadie (2021) each show, in different ways, why staggered rollout can distort naive two-way fixed-effects estimates when treatment effects vary across cohorts. I use that literature as a constraint on design, not as a ritual citation set.

8.4 Hedonic pricing and accessibility capitalisation

Rosen's hedonic framework matters because it gives the study a disciplined fallback. If accessibility changes are capitalised into prices, DLD transactions may still reveal something when corridor traffic data are unavailable. But the inference has to stay narrow: a price premium near a treated corridor may reflect perceived access gains; it is not direct evidence that traffic performance improved, and it is certainly not a complete welfare estimate.

8.5 CBA, SUMO, and decision translation

Cost-benefit analysis and simulation enter later in the chain and for different reasons. CBA can value heterogeneous effects, but it cannot manufacture effects that have not been identified. SUMO can support delay, idling, and emissions scenarios when observed microdata are incomplete, but I will not present simulation output as causal evidence. MCDA appears only at the translation stage, after estimates or scenarios have already been labelled by evidence level.

8.6 Literature gap matrix

Table 9. Literature-gap positioning matrix for study design.

Literature stream	What it explains	What it fails to prove	Role in this thesis
Smart mobility / ITS	Operational mechanisms and outcome categories.	Counterfactual welfare return of named Dubai interventions.	Defines outcomes and mechanisms.
Smart-city productivity	Digital infrastructure may affect economic performance.	Corridor-level treatment effects in Dubai.	Motivation only.
DiD / event studies	How to estimate treatment effects under a credible counterfactual.	No credibility without sharp treatment and valid controls.	Primary identification framework.
Hedonic pricing	How accessibility may capitalise into property values.	Direct traffic performance or total social welfare.	Secondary fallback evidence.
CBA / appraisal	How to monetise heterogeneous effects.	Causal effects when inputs are assumed.	Welfare-accounting layer.
SUMO / simulation	How traffic scenarios can be modelled.	Observed real-world treatment effects.	Scenario support only.

9. Data-Gated Research Design and Identification Strategy

9.1 Design philosophy

The design follows a strict evidence hierarchy. I first look for causal traffic effects. Only after that do I monetise observed effects. When direct traffic data are unavailable, I reduce the claim rather than force unlike evidence into the same box. Property prices and simulation outputs can still be informative, but they do different analytical work.

In operational terms, Level 1 means corridor- or intersection-level causal traffic evidence. Level 2 means secondary property-capitalisation evidence. Level 3 means scenario-based appraisal. I will carry that classification through the data dictionary, model tables, CBA workbook, and policy matrix so that the strength of the language always matches the strength of the evidence.

The design philosophy below is now translated into an executable workflow so that the reader can see the route from raw inputs to admissible claims.

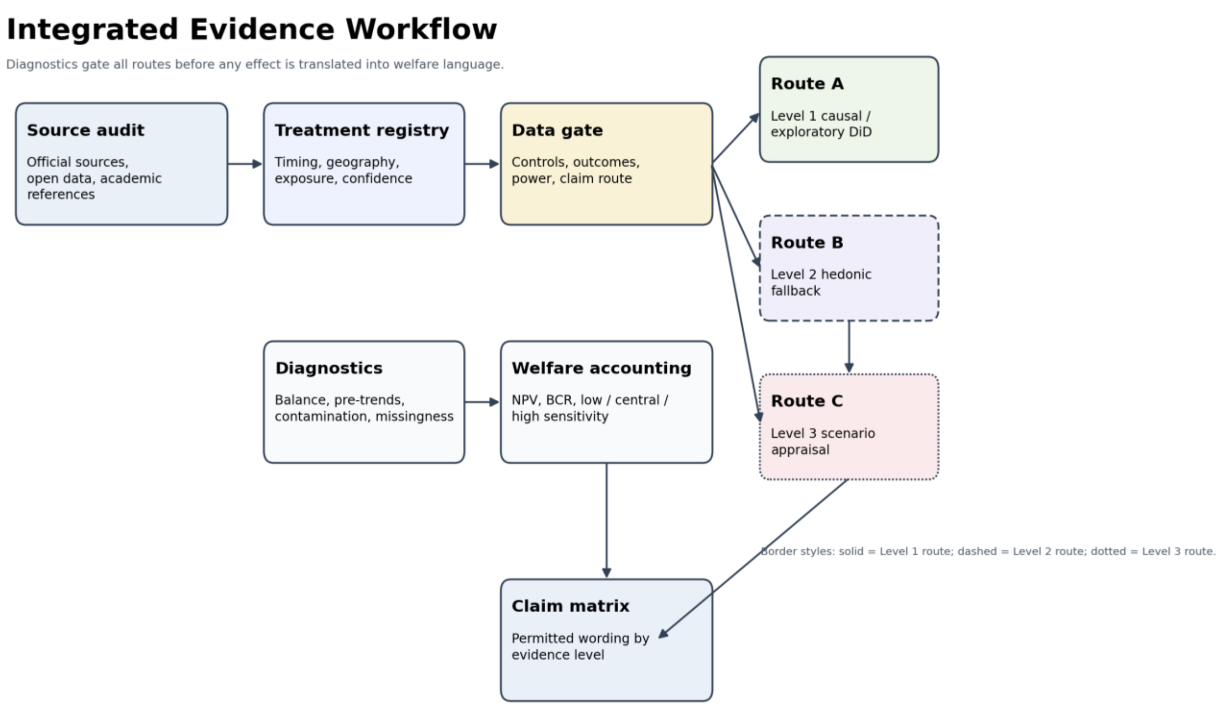

Figure 2. Integrated evidence workflow from source audit to claim matrix. Diagnostics gate all routes before any effect is translated into welfare language.

Because the thesis depends on multiple public and institutional sources, the data architecture must be explicit rather than implied.

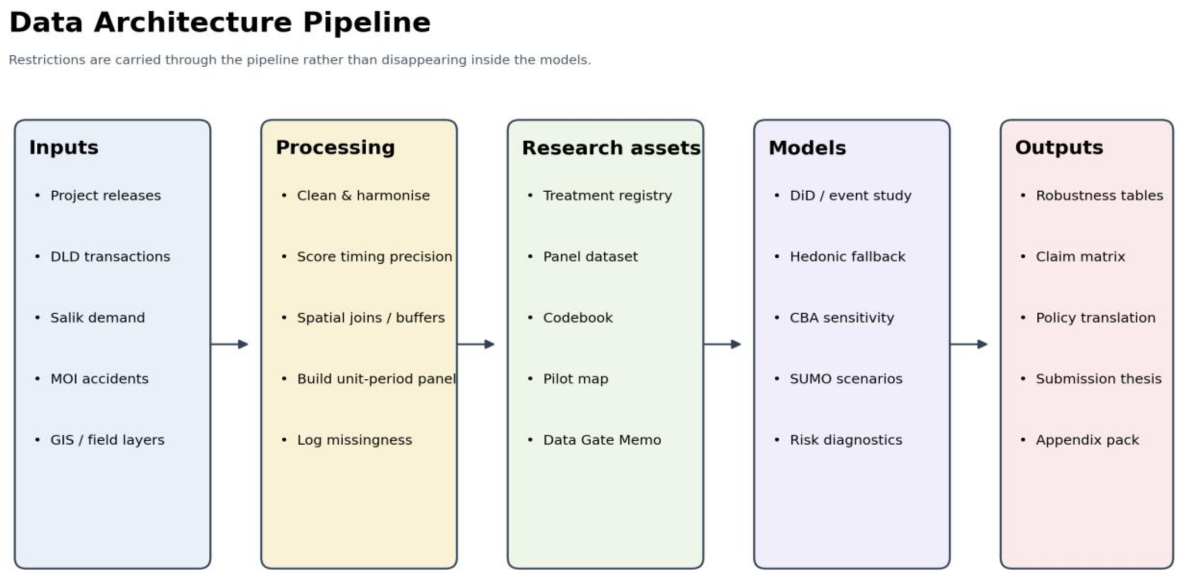

Figure 3. Data architecture pipeline from public inputs to model-ready assets and final outputs; data restrictions propagate to the final claim level.

9.2 Preliminary treatment registry

Table 10. Preliminary treatment registry based on currently visible evidence.

Intervention / data asset	Evidence visible now	Geography available now	Identification value	Main weakness
DITSC	AED 590m investment; ITS coverage from 11% to 60%.	System/network-level in public communication.	High policy relevance; medium treatment value.	Needs corridor or segment activation geography.
UTC-UX Fusion launch	AI, predictive analytics, and digital-twin signal-control rollout across major intersections.	Major-intersection/system-level public description.	High timing relevance; medium treatment value.	If rollout is too broad, untreated controls may vanish.
UTC-UX Phase 1 flow claim	Reported 16%-37% traffic-flow improvement across selected corridors.	Broad corridor references.	Useful as a mechanism clue and CBA scenario parameter.	Not counterfactual without raw outcome series.
DLD transaction data	Transactions, rents, projects, valuations,	Area/property-level fields; coordinates	Strong fallback value.	May lack exact geocoding or bulk

Intervention / data asset	Evidence visible now	Geography available now	Identification value	Main weakness
	property fields.	must be tested.		historical extraction.
Salik annual report	Revenue-generating trips and toll-demand indicators.	System/gate level depending on disclosure.	Useful demand proxy.	May be too aggregated for corridor DiD.
MOI accident data	Accident and injury categories.	Aggregate or emirate-level unless restricted data are granted.	Safety context.	May not support corridor-level safety effects.

9.3 Treatment coding protocol

Table 11. Treatment coding protocol for verified candidate units.

Coding element	Rule
Unit	Corridor, intersection, corridor-zone, community, or property buffer depending on data precision.
Treatment type	DITSC, UTC-UX, combined smart-traffic exposure, or context-only project.
Activation date	Month preferred; quarter accepted; year-only timing cannot support an event-study claim.
Geography precision	Intersection/corridor can support Level 1; community/buffer supports Level 2; system-only supports Level 3.
Exposure intensity	Binary, distance band, rollout cohort, or intensity score depending on geography.
Confidence score	High, medium, or low based on source quality and spatial/date precision.
Claim level	Every unit receives Level 1, Level 2, or Level 3 status before modelling.

9.4 Data gate and minimum thresholds

The hard data gate is the central feasibility mechanism of the proposal and is easier to audit as a decision tree than as a text-only threshold table.

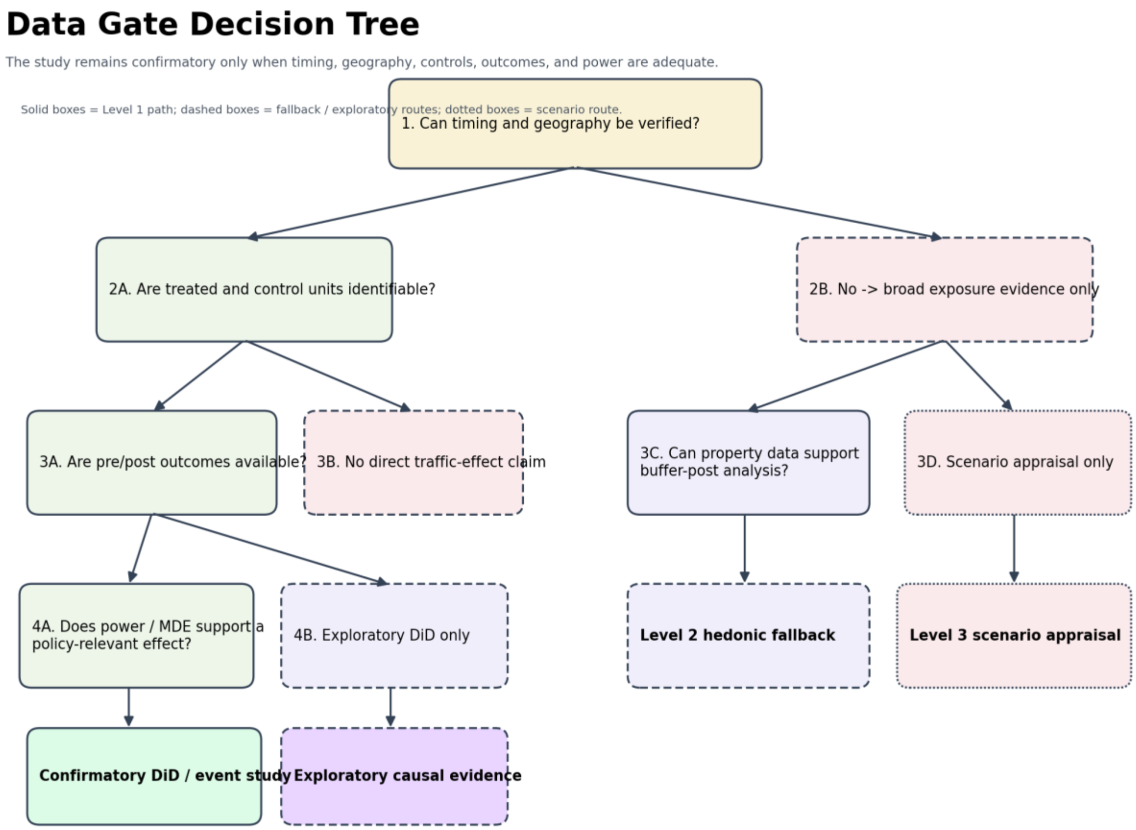

Figure 4. Data-gate decision tree showing when the study remains confirmatory, becomes exploratory, or downgrades to Level 2 or Level 3 evidence.

Table 12. Data-gate rule for confirmatory DiD eligibility.

Data gate rule
The DiD will proceed as confirmatory only if the available panel can detect a policy-relevant minimum effect. I use a provisional threshold of a 10 percent reduction in delay or travel-time variability, subject to observed pre-period variance. If the available panel cannot detect that scale of change, the DiD will be reported as exploratory rather than confirmatory.

Table 13. Minimum thresholds for confirmatory evidence and downgrade rules.

Requirement	Confirmatory threshold	If threshold fails
Treated units	At least 8 treated corridors, intersections, or zones unless synthetic control is justified.	Reclassify as exploratory or switch to Level 2/3.
Control units	At least 8 comparable untreated or later-treated units.	Use later-treated units or synthetic control; otherwise no causal traffic claim.
Treatment timing	Monthly or quarterly activation timing.	No event-study claim; use broad exposure only as secondary or scenario evidence.
Treatment geography	Bufferable corridor, intersection, or	System-level scenario appraisal only.

Requirement	Confirmatory threshold	If threshold fails
	community geometry.	
Pre/post periods	Preferred: 12 monthly periods before and after; minimum accepted only with power caution.	Classify as exploratory if the panel is too short.
Outcome data	Speed, delay, reliability, flow, safety, or credible proxy at unit-period level.	No direct traffic-effect claim.

Notes: The 10 percent effect threshold is provisional and subject to observed variance. Failing a threshold triggers exploratory reporting or methodological downgrade.

9.5 Estimand-data-method matrix

Table 14. Estimand-data-method mapping and permitted claim levels.

Estimand	Outcome	Unit	Treatment	Method	Claim allowed
ATT on traffic friction	Speed, delay, reliability, congestion proxy.	Corridor/intersection-month.	DITSC/UTC-UX exposure.	DiD/event study.	Causal if matching and pre-trends pass.
ATT on safety	Accident counts, injuries, severity proxy.	Corridor/intersection/area-period.	Smart-traffic exposure.	Count model plus DiD where possible.	Causal only with unit-level safety data.
Capitalisation effect	Log transaction price or price per square metre.	Property/community-month.	Buffer exposure.	Hedonic DiD.	Secondary evidence only.
Welfare return	NPV and BCR.	Corridor/project.	Observed or assumed effect size.	CBA sensitivity.	Appraisal; causal only when inputs are Level 1.
Scenario impact	Delay, idling, emissions.	Modelled corridor.	Assumed exposure.	SUMO scenarios.	Scenario evidence only.

Notes: ATT = average treatment effect on the treated; BCR = benefit-cost ratio. The last column states the strongest admissible claim for each design.

9.6 Sampling, matching, and controls

Controls will not be chosen by convenience. I will construct them in five steps:

1. Filter by road class and district/community type.
2. Identify untreated or later-treated units that share baseline network conditions with treated units.
3. Match on baseline congestion or demand proxy, Salik exposure, metro proximity, road hierarchy, development intensity, and pre-treatment trend distance.
4. Check balance with standardised mean differences and pre-period root mean squared prediction error.
5. Exclude units contaminated by major roadworks, metro projects, or system-wide treatment spillovers that cannot be modelled.

Table 15. Control-selection framework for the main empirical designs.

Control category	Examples	Purpose
Transport controls	Road hierarchy, baseline traffic, Salik exposure, metro proximity, roadworks.	Separate smart-traffic effects from network changes.
Economic controls	Interest rates, tourism cycles, property-market indicators, macro shocks.	Reduce bias in property-capitalisation analysis.
Spatial controls	CBD distance, land-use mix, density, accessibility, district fixed effects.	Account for location advantages.
Policy controls	Police campaigns, toll changes, metro expansion, major road projects.	Avoid attributing simultaneous policies to smart traffic.
Time controls	Month, quarter, year, seasonality, holidays.	Remove common shocks.

Notes: Control families shown here are a selection framework rather than a final regression specification.

9.7 Baseline DiD model

$$Yit = \alpha + \beta SmartTrafficit + \gamma Xit + \mu i + \tau t + \varepsilon it. \quad (1)$$

Where Y_it is the outcome for unit i in period t; α is a constant; SmartTraffic_it is smart-traffic exposure; X_it is the vector of time-varying controls; β and γ are coefficient terms; μ_i and τ_t are unit and time fixed effects; and ε_it is the error term. Outcome units should follow the underlying series, such as minutes of delay, speed-loss percentage, or a reliability index.

9.8 Event-study model and rejection rule

$$Yit = \alpha + \Sigma k\beta k1[t - Ti = k] + \gamma Xit + \mu i + \tau t + \varepsilon it. \quad (2)$$

Where Y_it is the outcome for unit i in period t; α is a constant; β_k is the coefficient for event time k; $1[t - T_i = k]$ is an indicator for being k periods from treatment; T_i is the treatment period for unit i; X_it is the vector of time-varying controls; γ is the associated coefficient vector; μ_i and τ_t are unit and time fixed effects; and ε_it is the error term. The period immediately before treatment is omitted. I will reject Equation (2) as confirmatory if the pre-treatment coefficients are jointly different from zero at the 10 percent level or if their magnitudes diverge systematically from the post-treatment effects.

9.9 Hedonic fallback model

$$ln(Pjct) = \alpha + \beta(Bufferc \times Postt) + \theta Zjct + \mu c + \tau t + \varepsilon jct. \qquad (3)$$

Where ln(P_jct) is the natural log of the transaction price for property j in community or buffer c at time t; Buffer_c marks treated proximity; Post_t marks the post-treatment period; Z_jct is the vector of observed property characteristics; β and θ are coefficient terms; μ_c and τ_t are community and time fixed effects; and ε_jct is the error term. Prices should be recorded in AED, or in AED per square metre if that specification is used.

9.10 CBA model

$$NPV = \Sigma t[(Benefitst - OPEXt)/(1 + r)t] - CAPEX. \qquad (4)$$

$$BCR = PV(Benefits)/PV(Costs). \qquad (5)$$

Where Benefits_t and OPEX_t are period-t benefits and operating costs in AED; r is the annual discount rate; CAPEX is the upfront capital cost in AED; PV(·) is the present-value operator; and BCR is unitless. Unless I can identify an official UAE public-sector appraisal rate, I will treat 6 percent, 8 percent, and 10 percent as scenario rates rather than official parameters. Every CBA table will separate observed inputs from assumed inputs, and reported flow improvements will be used only as scenario parameters unless counterfactual evidence is available.

Table 16. Parameter hierarchy for cost-benefit analysis inputs.

Parameter	Priority source	Fallback source	Status
CAPEX	RTA or contract value.	Public project benchmark range.	Observed or sensitivity.
OPEX	RTA annual or project report.	Percentage of CAPEX or comparable ITS benchmark.	Sensitivity parameter if unavailable.
Value of time	UAE/Dubai appraisal value if available.	World Bank or transport-economics benchmark.	Scenario parameter.
Accident cost	UAE or police valuation if available.	International willingness-to-pay benchmark.	Scenario parameter.
Emissions factor	Local transport/emissions source.	IEA or IPCC-compatible factor.	Scenario parameter.
Asset life	RTA documentation.	Infrastructure appraisal	Scenario parameter.

Parameter	Priority source	Fallback source	Status
		assumption.	

Notes: Table distinguishes preferred data sources from fallback assumptions for cost-benefit analysis.

9.11 Conditions that invalidate causal claims

Table 17. Conditions that invalidate causal claims.

Invalidation condition	Consequence
Treatment timing cannot be verified.	No event-study or confirmatory DiD.
Treatment geography is system-level only.	No corridor or intersection causal claim.
UTC-UX rollout leaves no credible controls.	Use cohort, intensity, synthetic control, or downgrade claims.
Pre-trends fail.	Report as descriptive or exploratory only.
Outcomes are annual or citywide only.	No unit-level treatment-effect claim.
Major confounders cannot be excluded.	Downgrade to sensitivity or descriptive reporting.
CBA uses assumed effects only.	Report scenario appraisal, not verified welfare return.

Notes: These are claim-invalidation rules; they do not imply total project failure, only a lower admissible evidence level.

9.12 QGIS, SUMO, and MCDA discipline

QGIS will be used for concrete spatial tasks: mapping exposure, constructing buffers, linking units, and testing spatial decay. SUMO will be reserved for scenario analysis when observed microdata are incomplete; I will not describe those outputs as causal evidence. MCDA remains at the policy-translation stage, where rankings are reported alongside uncertainty and evidence level rather than as stand-alone truth claims.

10. Contribution, Policy Value, and Expected Evidence Outputs

10.1 Five-year value

Five years from now, the value of this thesis should not rest on a broad claim that ‘smart mobility works.’ That would be too vague to be useful. A better outcome is a tighter one: researchers and policy analysts would have a reproducible way to decide what can be learned from smart-traffic investments when agency microdata are partial, delayed, or restricted. The framework matters because it tells users when a causal claim is justified, when property evidence is only secondary, and when CBA is no more than scenario appraisal.

10.2 Expected outputs

11. Treatment registry with source, timing, geography, confidence, and claim-level classification.
12. Data Gate Memo by Month 3.
13. Primary or fallback dataset with data dictionary and missingness log.
14. Econometric output classified as causal, exploratory, secondary, or scenario-based.
15. CBA workbook with low, central, and high parameter scenarios.
16. Claim-aware policy-translation matrix.

16.1 Non-overclaiming rule

Positive, null, and negative results are all informative here, but they do not carry the same meaning. A positive Level 1 result would support measurable traffic-performance gains. A null Level 1 result would indicate that the observable improvement does not survive noise or counterfactual uncertainty. A Level 2 result would speak only to local property-capitalisation. A Level 3 result would remain a scenario exercise. I will not turn a weaker evidence class into a stronger claim by wording alone.

17. Execution Plan: Timeline, Budget, Ethics, and Risk Controls

17.1 First twelve weeks: survival plan

The first twelve weeks constitute the feasibility stress test of the project; the sequence is therefore shown as a true Gantt rather than a static table.

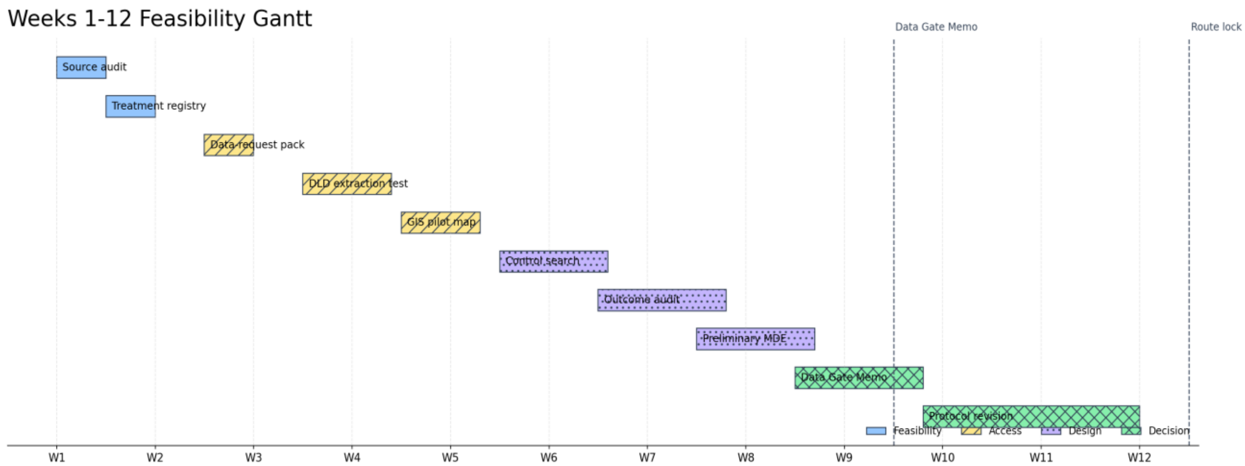

Figure 5. Weeks 1-12 feasibility Gantt showing the critical sequence from source audit to Data Gate Memo and route lock.

Table 18. Weeks 1-12 feasibility plan, deliverables, and go/no-go dependencies.

Week	Task	Deliverable	Go / No-Go dependency
W1	Audit official sources for DITSC, UTC-UX, DLD, Salik, MOI, and ADU.	01_source_log.xlsx	At least 10 sources classified by use and limitation.
W2	Build treatment registry v0.	02_treatment_registry.csv	Candidate units listed or impossibility documented.
W3	Prepare data-request package.	Request templates	RTA, DLD, MOI, and Salik requests ready or sent.
W4	Run DLD extraction test.	04_DLD_sample_transactions.xlsx	If fewer than 1,000 usable rows are available, property-level hedonic design downgrades to community-level.
W5	Create GIS exposure prototype.	05_QGIS_pilot_map.png	If geography is system-level only, corridor DiD is abandoned.
W6	Search control units.	Candidate-control sheet	If no credible controls exist, use cohort/intensity design or downgrade.
W7	Audit outcome availability.	Outcome memo	If outcomes are citywide only, no Level 1 claim.
W8	Run preliminary MDE logic.	MDE note	If MDE exceeds a policy-

Week	Task	Deliverable	Go / No-Go dependency
			relevant threshold, DiD becomes exploratory.
W9	Draft Data Gate Memo.	03_data_gate_memo.pdf	Choose Level 1, exploratory, Level 2, or Level 3 route.
W10-W12	Revise protocol with supervisor feedback if available.	Revised protocol	Proceed only with the evidence route that survived.

Notes: W = week.

17.2 Twelve-month timeline

The annual work plan is also rendered as a Gantt so that sequencing, overlap, and critical milestones remain visible in one scan.

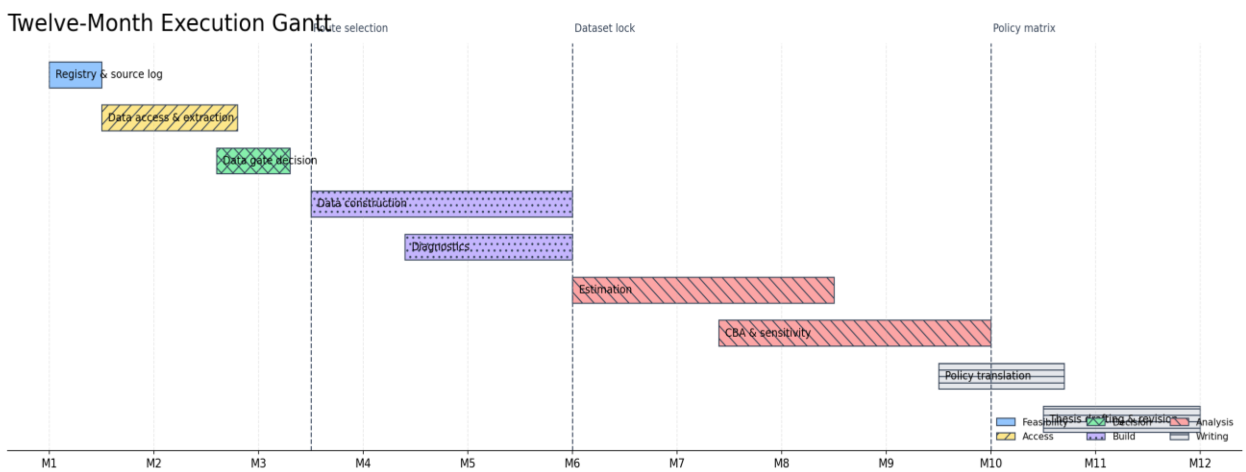

Figure 6. Twelve-month execution Gantt showing route selection, dataset lock, estimation, appraisal, and final writing milestones.

Table 19. Twelve-month work plan, phases, and deliverables.

Month	Phase	Core task	Deliverable
M1	Feasibility start	Source audit and treatment registry v0.	Registry v0 and source log.
M2	Data access	Requests, DLD extraction test, GIS prototype.	Data-access package and sample dataset.
M3	Data gate	MDE check and route decision.	Data Gate Memo.
M4	Data construction	Build primary or fallback panel.	Dataset v1.
M5	Diagnostics	Matching, balance, pre-trends, missingness.	Diagnostics report.
M6	Dataset lock	Freeze model-ready dataset.	Dataset v2 and codebook.

Month	Phase	Core task	Deliverable
M7-M8	Estimation	DiD/event-study or fallback models.	Model output and robustness tables.
M8-M10	Appraisal	CBA and sensitivity scenarios.	CBA workbook.
M10-M11	Translation	Claim-aware matrix and interpretation.	Policy matrix.
M11-M12	Writing	Final thesis drafting and revision.	Submission-ready thesis.

Notes: M = month. Timings after Month 3 depend on the route chosen in the Data Gate Memo.

17.3 Deliverables and acceptance criteria

Table 20. Deliverables and acceptance criteria for the proposed project.

Deliverable	Acceptance criterion
01_source_log.xlsx	Each source records variable found, granularity, use, limitation, access status, and claim level.
02_treatment_registry.csv	Each unit has source, date precision, geography precision, exposure level, and confidence score.
03_data_gate_memo.pdf	Explicitly chooses Level 1, exploratory, Level 2, or Level 3 route with reasons.
04_panel_dataset_v1.csv	Contains unit identifiers, period identifiers, treatment variable, outcomes, controls, and missingness flags.
05_codebook.xlsx	Defines variables, formulas, source, unit, frequency, and transformation.
06_pretrend_balance_report.docx	Shows matching balance, pre-trends, contamination checks, and rejected units.
07_cba_workbook.xlsx	Labels observed versus assumed inputs and reports low, central, and high scenarios.
08_claim_classification_matrix.xlsx	Maps every result to permitted and forbidden wording.

Notes: Acceptance criteria function as proposal-management benchmarks for each deliverable.

17.4 Indicative budget

The budget becomes more legible when fixed and contingent cost ranges are separated visually.

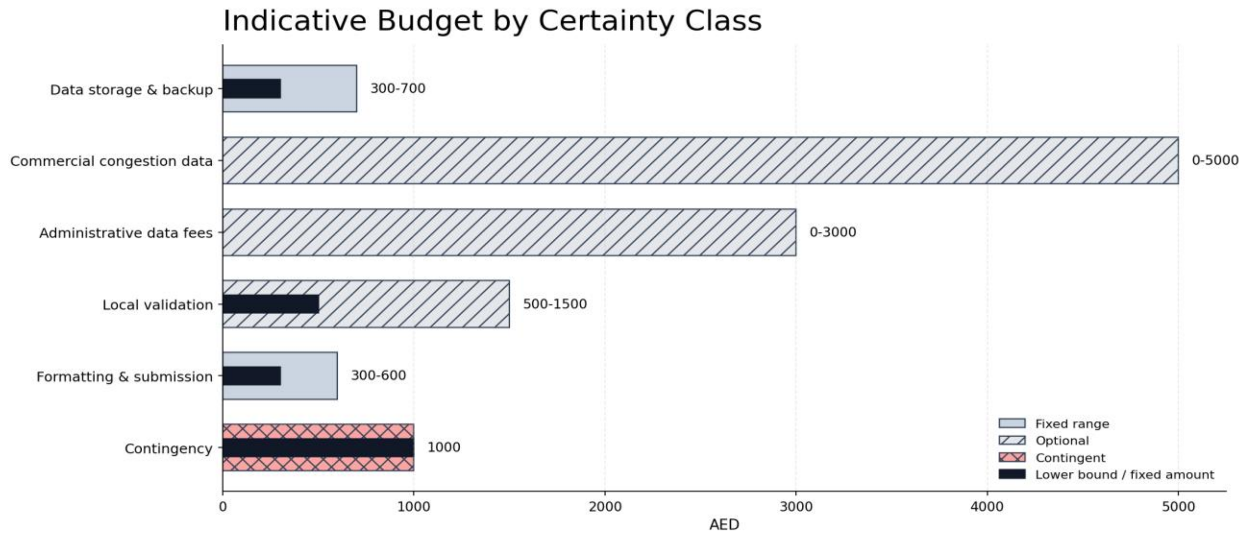

Figure 7. Indicative budget by certainty class, separating fixed ranges, optional expenditures, and contingent costs.

Table 21. Indicative 12-month project budget (AED).

Item	Estimated cost (AED)	Justification
Public datasets	0	Core design starts with official open data.
QGIS and SUMO	0	Open-source spatial and simulation tools.
Data storage and backup	300-700	Version control, backups, secure storage.
Optional commercial congestion data	0-5,000	Only if INRIX, TomTom, or xMap access is required and affordable.
Optional administrative data fees	0-3,000	Possible institutional data-access charges.
Local corridor validation	500-1,500	Optional field observation and map validation.
Formatting and submission	300-600	Final document preparation.
Contingency	1,000	Unplanned data, access, or submission costs.
Data-access failure cost	Methodological downgrade	If Level 1 data fail, the study changes claim level rather than buying credibility.

Notes: Ranges indicate planning estimates. Costs can be fixed, optional, or contingent depending on data access needs.

17.5 Ethics and data governance

The study will use public, institutional, and secondary data. It will not collect personally identifiable driver data. Restricted traffic or accident data will be stored and reported under the relevant data-sharing agreement. The thesis will not publish sensitive traffic-control locations, raw geocoded accident records, or operational details that could create security risk. If interviews or surveys are later added, ADU ethics approval will be sought before any contact with participants.

17.6 Risk register

The risk register is complemented with a likelihood-impact heatmap so that the committee can identify the principal failure modes immediately.

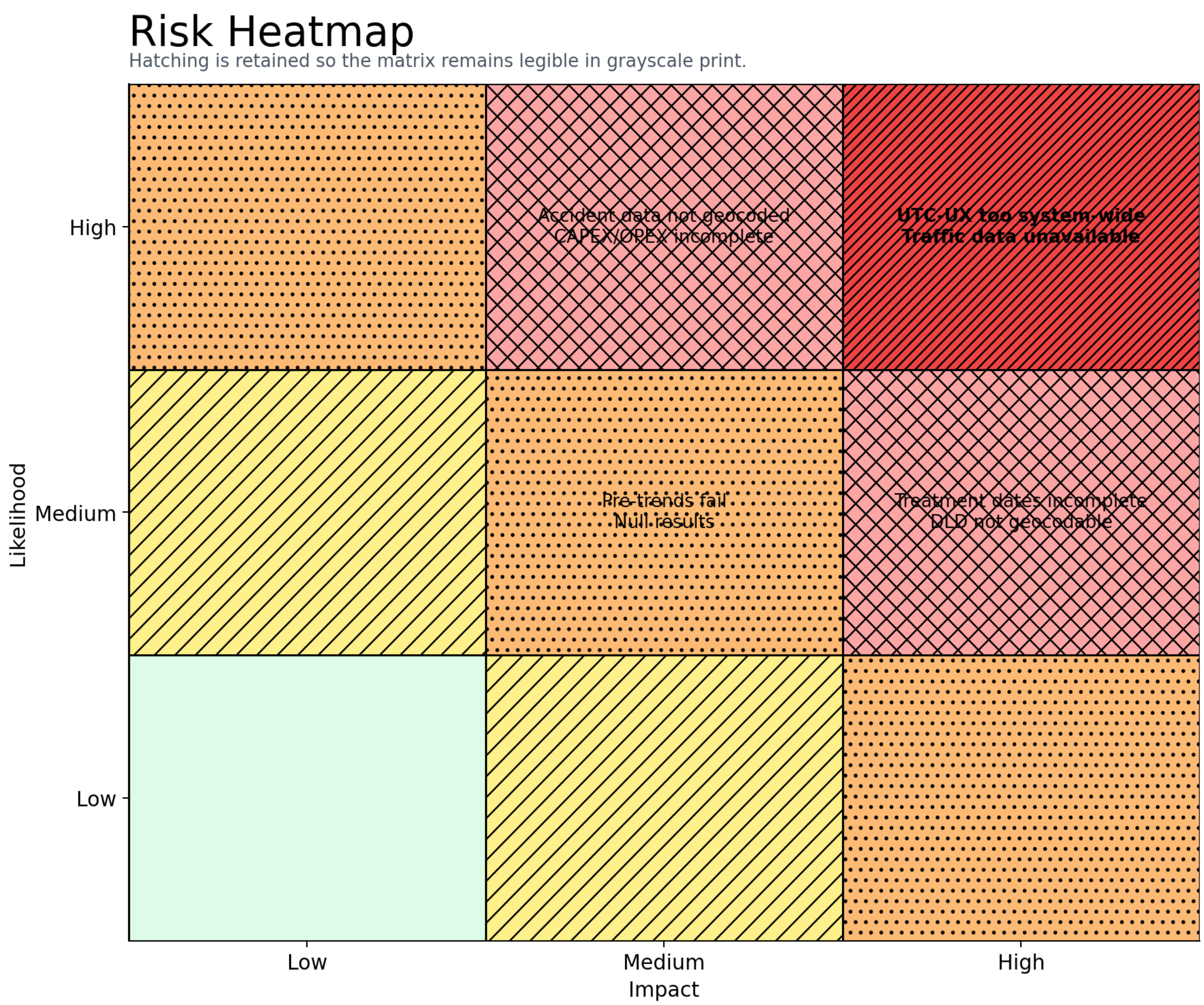

Figure 8. Risk heatmap showing the principal downgrade-triggering risks by likelihood and impact.

Table 22. Risk register with mitigation rules.

Risk	Likelihood	Impact	Mitigation
Treatment dates incomplete.	Medium	High	Use confidence scoring; exclude ambiguous units; downgrade if needed.
UTC-UX too system-wide.	High	High	Use rollout timing, exposure intensity, later-treated units, or downgrade.
Corridor traffic data unavailable.	High	High	Switch to hedonic fallback and scenario CBA; no causal traffic claim.
DLD data not geocodable enough.	Medium	High	Use community-level analysis or downgrade to descriptive property context.
Accident data not geocoded.	High	Medium	Use safety context and scenario valuation only.
CAPEX/OPEX incomplete.	Medium	High	Use source hierarchy and sensitivity ranges.
Pre-trends fail.	Medium	High	Reject confirmatory DiD and report exploratory findings only.
Results are null.	Medium	Medium	Report as data-constrained evidence; do not reinterpret as technology failure.

Notes: Likelihood and impact are ordinal categories used for project-risk management.

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